

5. Write a C program for generalization of the Caesar cipher, known as the affine Caesar cipher, has the following form: For each plaintext letter p , substitute the ciphertext letter C : $C = E([a, b], p) = (ap + b) \bmod 26$ A basic requirement of any encryption algorithm is that it be one-to-one. That is, if $p \neq q$, then $E(k, p) \neq E(k, q)$. Otherwise, decryption is impossible, because more than one plaintext character maps into the same ciphertext character. The affine Caesar cipher is not one-to-one for all values of a . For example, for $a = 2$ and $b = 3$, then $E([a, b], 0) = E([a, b], 13) = 3$.

a. Are there any limitations on the value of b ?

b. Determine which values of a are not allowed.

Aim:

To write a C program to implement the Affine Caesar cipher using the formula:

$$C = (ap + b) \bmod 26$$

and determine the limitations on the values of a and b .

Algorithm:

1. Start the program.
2. Read the plaintext, and values of a and b .
3. Check whether $\gcd(a, 26) = 1$.
4. Convert each plaintext letter into a number from 0 to 25.
5. Apply the formula $C = (a \times P + b) \bmod 26$.
6. Convert the calculated values back into letters.
7. Display the ciphertext and stop the program.

Program:



```
main.c Run ▶  
1  #include <stdio.h>  
2  #include <string.h>  
3  #include <ctype.h>  
4  
5  int gcd(int a, int b)  
6  {  
7      while (b != 0)  
8      {  
9          int temp = b;  
10         b = a % b;  
11         a = temp;  
12     }  
13     return a;  
14 }  
15  
16 int main()  
17 {  
18     char plaintext[100], ciphertext[100];  
19     int a, b, i;  
20  
21     printf("Enter plaintext: ");
```



```
main.c Run ▶  
38  
39     for (i = 0; plaintext[i] != '\0'; i++)  
40     {  
41         if (isalpha(plaintext[i]))  
42         {  
43             int p = toupper(plaintext[i]) - 'A';  
44  
45             ciphertext[i] = (a * p + b) % 26 + 'A';  
46         }  
47         else  
48         {  
49             ciphertext[i] = plaintext[i];  
50         }  
51     }  
52  
53     ciphertext[i] = '\0';  
54  
55     printf("Ciphertext: %s", ciphertext);  
56  
57     return 0;  
58 }
```

Output:

Output

```
Enter plaintext: HELLO  
Enter value of a: 5  
Enter value of b: 8  
Ciphertext: RCLLA
```

Result:

Thus, the Affine Caesar cipher was successfully implemented in C. It was determined that b can have any value from 0–25, while a must be relatively prime to 26 for the encryption to be one-to-one.